

William Yen

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EDUCATION

University of Texas at Austin

Austin, TX

B.S. in Computer Science, Minor in Business Administration

Expected May 2028 | GPA: 3.66

- **Relevant Coursework:** Data Structures, Algorithms & Complexity, Operating Systems, Computer Systems, Computer Architecture, Computer Vision, Linear Algebra, Probability & Statistics

EXPERIENCE

Software Engineer Intern

Incoming Oct. 2026

PwC — Tax Innovation & Delivery Experience

- Joining the Tax Innovation & Delivery Experience team to build AI tooling that modernizes tax workflows

Software Developer Intern

May 2026 – Aug. 2026

IBM (zAIOps)

Poughkeepsie, NY

- Owned the ServiceNow agent for IBM's Z-mainframe LLM assistant end to end — design, implementation, testing, and release — orchestrating 19 Python tools across distributed LangGraph services
- Debugged a systematic routing failure by analyzing production failure traces, then rewrote the prompt routing table across 14 workflows to raise accuracy from 33% to 97% (71/73 cases)
- Profiled a bottleneck to an N+1 query pattern in a graph traversal and rewrote it with two-phase batch fetching, cutting latency 25–45% and API calls up to 47%
- Automated release validation with a 73-case Promptfoo test harness using LLM-as-judge grading, cutting regression cycles from 2 min to 5–10 s and catching correctness defects before merge

Student Technician

Sep. 2025 – Present

UT Austin Enterprise Technologies

Austin, TX

- Designed, deployed, and maintain production RESTful APIs in Python over a MySQL backend serving 70,000+ users
- Engineered a Java API proxy over Wi-Fi client data (MACs, hostnames, user IDs) from 40+ buildings
- Cut manual network configuration time 90% through clean, maintainable object-oriented design

AI/ML Student Researcher

May 2024 – Aug. 2024

University of Texas at Dallas

Dallas, TX

- Trained image classification models in PyTorch and TensorFlow to 85% accuracy on a 50,000-image dataset
- Accelerated batch inference from 2 hours to 3 minutes (40×) via architecture optimization and parallel loading

PROJECTS

SyllaSync — AI Calendar Tool | *TypeScript, Next.js, React, Supabase (PostgreSQL), Groq, Gemini, Netlify*

- Built, deployed, and maintained a full-stack web application used by 100+ students that turns an uploaded syllabus PDF into reviewable assignment and exam events, exported to Google Calendar via OAuth or as an .ics file
- Improved reliability and cost by filtering date-free pages to cut LLM token spend, with an automatic Gemini fallback on Groq rate limits

PassForm — Volleyball Form Analyzer | *Python, YOLOv8, MediaPipe, OpenCV, NumPy*

- Debugged a fine-tuned ball detector that had learned ceiling lights, proving it with a transplant test (118/118 → 12/118), then swapped in stock COCO YOLOv8 to correct a scoring window running 16 frames late
- Rearchitected pose estimation as YOLO-pose multi-person tracking feeding per-athlete MediaPipe measurement, cutting runtime from 77s to 25s across 34 test cases

Slack Semantic Search (MCP Server) | *Python, FastMCP, OpenAI, WatsonX, Slack API*

- Built a server exposing 4 MCP tools for semantic search over Slack history with cited source messages; tied 5th of 93 teams at IBM's internal hackathon
- Migrated the embedding backend from WatsonX to OpenAI behind a stable interface with no change to the exposed tools, validated by a 59-test, 20-query automated eval harness

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, TypeScript, SQL

Software Development: Data structures & algorithms, object-oriented design, RESTful APIs, Node.js, React.js, Next.js, PostgreSQL, MySQL, NoSQL, OAuth

Testing & Developer Tools: Automated testing, debugging & profiling, performance tuning, Git/GitHub, CI/CD, Docker, AWS, Agile, AI-assisted development (Claude Code), LLM evaluation & benchmarking

AI & ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, YOLO, MediaPipe, LangGraph, LangChain, RAG & information retrieval, ChromaDB